Mohammad Asgar Khan Dalwai

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PROFESSIONAL SUMMARY

Data Science master's student at Arizona State University (GPA 4.0) with hands-on deep learning experience in computer vision and speech. Built a CNN eye disease classifier using MobileNetV2 transfer learning in TensorFlow/Keras. Fine-tuned a 244M-parameter Whisper model in PyTorch, cutting word error rate by 6.2 points on unseen speakers, with confidence intervals and multi-seed runs. Two published papers on computer vision applications. Experienced in OpenCV preprocessing, standard evaluation metrics and SHAP explainability.

EDUCATION

Master of Science in Data Science May 2027
Arizona state university, Tempe AZ GPA 4.0
RelevantCourses: Analyzing Big Data, Statistical Machine Learning, Statistics for Data Analysts, DataProcessingat Scale, Data Mining, Data visualization, Advanced Database Management System, Advanced Computer Network Security

Bachelor of Technology in computer Science and Business System June 2024
Bharati Vidyapeeth (DU) DET, Mumbai India GPA 3.4
Relevant Courses: Data Structures and Algorithms, Computer Networks, Database Management Systems, Machine Learning, Internet of Things, Operating Systems, Information Security

SKILLS

Programming Languages: Python, SQL, Java

Computer Vision: OpenCV, image preprocessing (resizing, normalization, augmentation), MobileNetV2, facial recognition, medical image classification

Deep Learning: PyTorch (Hugging Face Transformers), TensorFlow/Keras, transfer learning, fine-tuning, data augmentation, mixed-precision GPU training.

Evaluation: Accuracy, Precision, Recall, F1-score, word error rate, bootstrap confidence intervals, multi-seed experiments, error analysis

Tools: Git, GitHub Actions, Docker, FastAPI, Flask, Jupyter, Google Colab

PROFESSIONAL EXPERIENCE

Artificial Intelligence Trainee Feb 2022 – April 2022
LanguifyAi, Mumbai india

- Built an Automatic Speech Recognition (ASR) system combining acoustic and language modeling to recognize Indian-accented English speech.
- Implemented MFCC feature extraction, HMM decomposition, and ANN-based deep learning to integrate acoustic modeling, language modeling, and data preprocessing, enhancing overall transcription accuracy.
- Reduced WER to ~18% using custom dataset consisting of 500+ audio recordings by training and evaluating deep learning models on this data, which increased accuracy in speech recognition by ~15%.

ACADEMIC PROJECTS

Multiple Eye Disease Detection Using machine learning Fall 2023
Bharati Vidyapeeth (DU) DET

- Built a CNN classifier that sorts retinal images into 5 classes (cataract, diabetic retinopathy, glaucoma, normal, other), combining 4 public datasets (Kaggle, IDRiD, ORIGA, DRIVE).
- Fine-tuned a pre-trained MobileNetV2 with transfer learning in TensorFlow/Keras on 1000+ retinal images.
- Built the detection pipeline using MobileNetV2 CNN with transfer learning, fine-tuning pre-trained weights on 1,000+ retinal images; applied data augmentation (resize, normalize, flip) to correct class imbalance between healthy and diseased samples, improving generalization by ~18%.
- Rebuilt the Flask app for deployment with a Docker container, input validation, 14 automated tests and GitHub Actions CI.

Fine-Tuning Whisper for Accented English Speech Recognition Fall 2026
Arizona State University

- Fine-tuned OpenAI Whisper-small (244M parameters) in PyTorch on 15.9 hours of accented English conversations (EdAcc). Cut word error rate from 22.5% to 16.3% on 28 unseen test speakers (95% CI for the gain: 4.2 to 8.7 points).
- Found that fine-tuning made the model worse on ordinary read speech, then fixed it by mixing 15% LibriSpeech clips back into training. This cut the damage by more than half.
- Validated results with a locked test set, 3 random seeds, speaker-level bootstrap and manual error analysis. Showed that 2 to 3 points of the gain hold even after controlling for differences in transcription style.
- Shipped an int8 model (253 MB) as a FastAPI service in Docker on a public Hugging Face Space, with 47 automated tests.

PUBLICATIONS

Automated Attendance Management System Using Facial Recognition
IAES International Journal of Artificial Intelligence (IJ-AI), Vol. 13, No 1, March 2024 Part II

Multiple Eye disease Detection Using Machine learning
International Conference on Intelligent Computing and Communication Techniques (ICICCT-2024), held from 28th to 29th June, 2024